

DH Part 2

EGR 445/545

GVSU

Plan for today

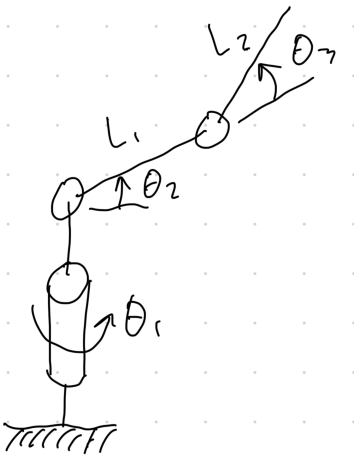
- review
- more DH
 - figure 3.15 and eqn. 3.6
- project time

DH Review

- what did we learn last time?
 - why are we going to use the DH convention?

Where does O_1 go?

... and why can't it go at the base?



Order of Multiplication and Eqn. 3.6

$${}^{i-1}_i \mathbf{T} = HT_x(\alpha_{i-1}, x = a_{i-1}) \cdot HT_z(\theta_i, z = d_i)$$

- matrix multiplication is not commutative
 - therefore α_{i-1} and a_{i-1} must come before θ_i and d_i

DH Part 2

Two Main Phases to the DH Convention

1. assign the $\hat{Z}$ axes
2. assign the origins and the $\hat{X}$ axes
 - the relationship between $\hat{Z}_i$ and $\hat{Z}_{i-1}$ determines where O_{i-1} goes and where $\hat{X}_{i-1}$ points

Three Cases for Z Axes

1. Z_{i-1} and Z_i are parallel
2. Z_{i-1} and Z_i intersect
3. Z_{i-1} and Z_i are not parallel and do not intersect

Case 1: Parallel Z Axes

- this is the only case where you have some freedom in placing an origin
 - but your freedom is limited to sliding the origin along Z_{i-1}
- X_{i-1} still goes along the common perpendicular line between Z_i and Z_{i-1}
 - but now there are infinitely many common perpendiculars

Case 2: Intersecting Z Axes

- if two consecutive Z axes intersect:
 - the origin for frame $i - 1$ must go at the intersection
 - X_{i-1} must be perpendicular to the plane formed by Z_{i-1} and Z_i
- **why?**

Case 3: Non-parallel and Non-intersecting

- there is only one common perpendicular line
- X_{i-1} must go along the common perpendicular
- the origin for frame $i - 1$ must go at the intersection of the common perpendicular and Z_{i-1}
 - i.e. X_{i-1} and Z_{i-1} intersect at the origin of frame $i - 1$
- this case is shown in Fig. 3.15

i vs. $i - 1$

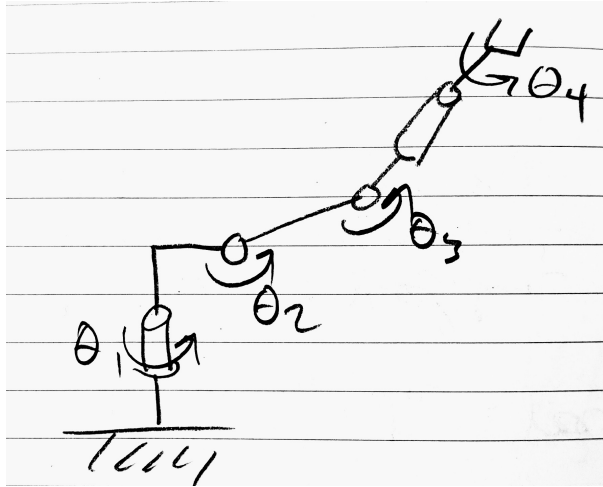
- we start at frame $\{i - 1\}$
- α_{i-1} and a_{i-1} refer to X_{i-1}
- θ_i and d_i refer to Z_i
- the parameter subscripts are chosen to agree with the axis subscripts
 - this will eventually feel less weird

DH Guiding Principle/Verification Question

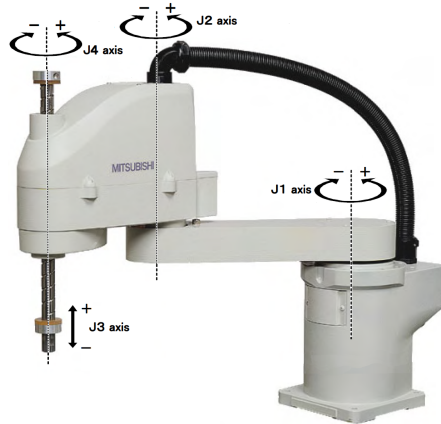
It **must** be possible to go from frame $i - 1$ to frame i by following these steps **in the given order**:

1. rotate by α_{i-1}
2. translate by a_{i-1}
3. rotate by θ_i
4. translate by d_i

Frame Attachment Example



SCARA Example



By Mitsubishi Electric Automation, Inc. 500 Corporate Woods Pkwy, Vernon Hills, IL 60061 <https://commons.wikimedia.org/w/index.php?curid=133585140>

Modified SCARA 1: RRP Planar

Modified SCARA 2: 3R with Rotated Third Joint

Big Ideas for Chapter 3

- any robot can be modeled with a series of HT matrices:

$${}^0P_{tip} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_4\mathbf{T} \cdot {}^4_5\mathbf{T} \cdot {}^5_6\mathbf{T} \cdot {}^6P_{tip}$$

- any robot link can be modeled using an HT matrix according to eqn. 3.6
 - two rotations and two translations
 - first, rotation and translation involving X_{i-1}
 - then rotation and translation involving Z_i

Key Points for Chapter 3

- Figure 3.5/3.15
- Derivation of eqn. 3.6
- DH parameter definitions
- DH Frame attachment procedure
 - the relationship between $\hat{Z}_i$ and $\hat{Z}_{i-1}$ determines where O_{i-1} goes and where $\hat{X}_{i-1}$ points

Figure 3.15

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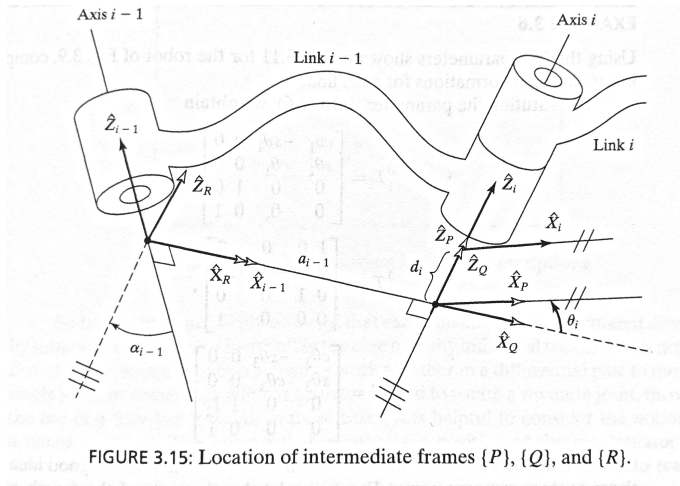
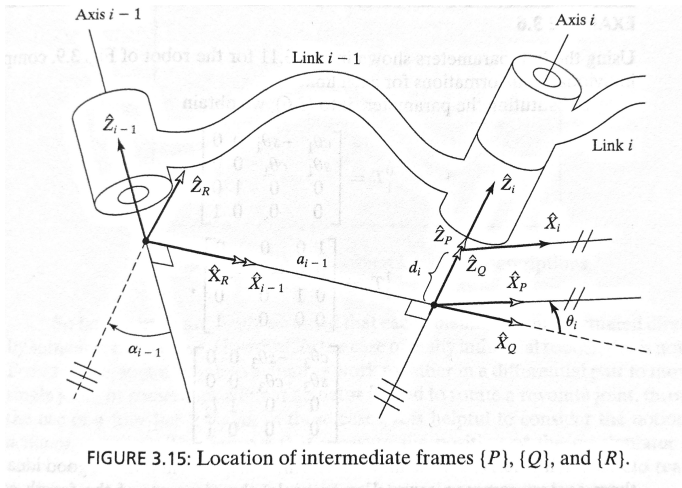


FIGURE 3.15: Location of intermediate frames $\{P\}$, $\{Q\}$, and $\{R\}$.

Figure 3.15 and Eqn. 3.6

$${}^{i-1}_i\mathbf{T} = {}^{i-1}_R\mathbf{T} \cdot {}^R_Q\mathbf{T} \cdot {}^Q_P\mathbf{T} \cdot {}^P_i\mathbf{T}$$



Eqn. 3.6

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- `part1 = robotics.HTx(alpha, x=a)`
- `part2 = robotics.HTz(theta, z=d)`
- `T = np.dot(part1, part2)`

Part 1 of Eqn. 3.6

$$\text{part1} = \begin{bmatrix} 1 & 0 & 0 & a_{i-1} \\ 0 & \cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & 0 \\ 0 & \sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- rotation by angle α_{i-1} about X
- translation by a_{i-1} along X

Part 2 of Eqn. 3.6

$$\text{part2} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & 0 \\ \sin(\theta_i) & \cos(\theta_i) & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- rotation by angle θ_i about Z
- translation by d_i along Z

np.dot(part1, part2)

$${}^{i-1}_iT = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & a_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\alpha_{i-1})\cos(\theta_i) & -\sin(\alpha_{i-1}) & -d_i\sin(\alpha_{i-1}) \\ \sin(\alpha_{i-1})\sin(\theta_i) & \sin(\alpha_{i-1})\cos(\theta_i) & \cos(\alpha_{i-1}) & d_i\cos(\alpha_{i-1}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

DH Frame Attachment Procedure

Attachment Procedure - Book

Summary of link-frame attachment procedure

The following is a summary of the procedure to follow when faced with a new mechanism, in order to properly attach the link frames:

1. Identify the joint axes and imagine (or draw) infinite lines along them. For steps 2 through 5 below, consider two of these neighboring lines (at axes i and $i + 1$).
2. Identify the common perpendicular between them, or point of intersection. At the point of intersection, or at the point where the common perpendicular meets the i th axis, assign the link-frame origin.
3. Assign the $\hat{Z}_i$ axis pointing along the i th joint axis.
4. Assign the $\hat{X}_i$ axis pointing along the common perpendicular, or, if the axes intersect, assign $\hat{X}_i$ to be normal to the plane containing the two axes.
5. Assign the $\hat{Y}_i$ axis to complete a right-hand coordinate system.
6. Assign $\{0\}$ to match $\{1\}$ when the first joint variable is zero. For $\{N\}$, choose an origin location and $\hat{X}_N$ direction freely, but generally so as to cause as many linkage parameters as possible to become zero.

Frame Attachment Gotchas

- steps 1 and 3 are done together and are straight forward
- steps 2 and 4 trip up students
- step 2: find the origins
- step 4: assign the X axes

Origin Choices

- most common mistake: thinking you have freedom to place the origin where you want
- you only really have freedom if two consecutive Z axes are parallel

Frame Attachment Summary

- steps 1 & 3: identify the joint axes and place Z_i along them
- steps 2 & 4: identify the common perpendiculars for each pair of Z axes, Z_{i-1} and Z_i
 - depends on the Z axis case (next slide)
 - X_{i-1} along the common perpendicular for Z_{i-1} and Z_i
- step 5 (skip): Y_i to complete RH coordinate system
- step 6: Decide on frames $\{0\}$ and $\{N\}$